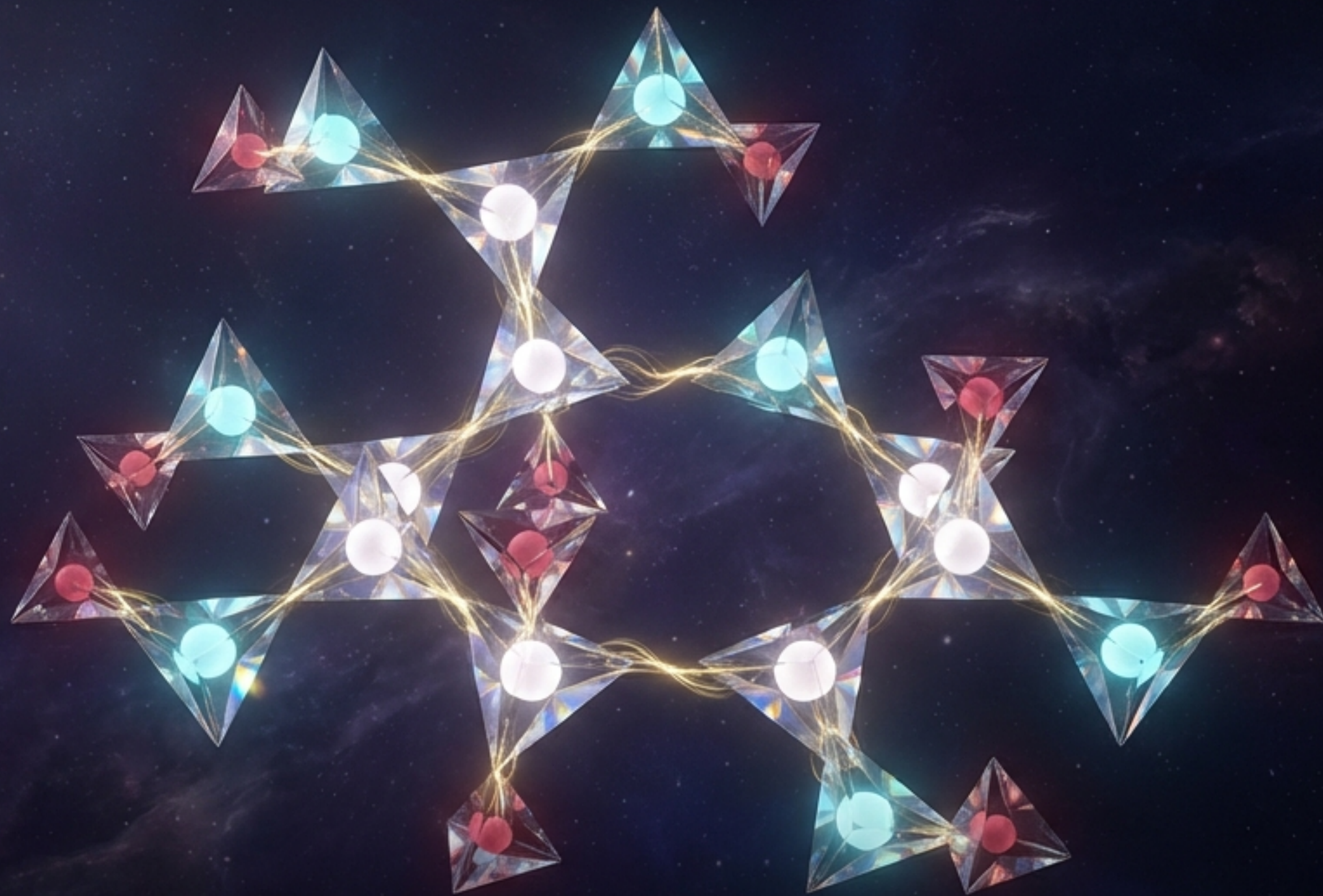


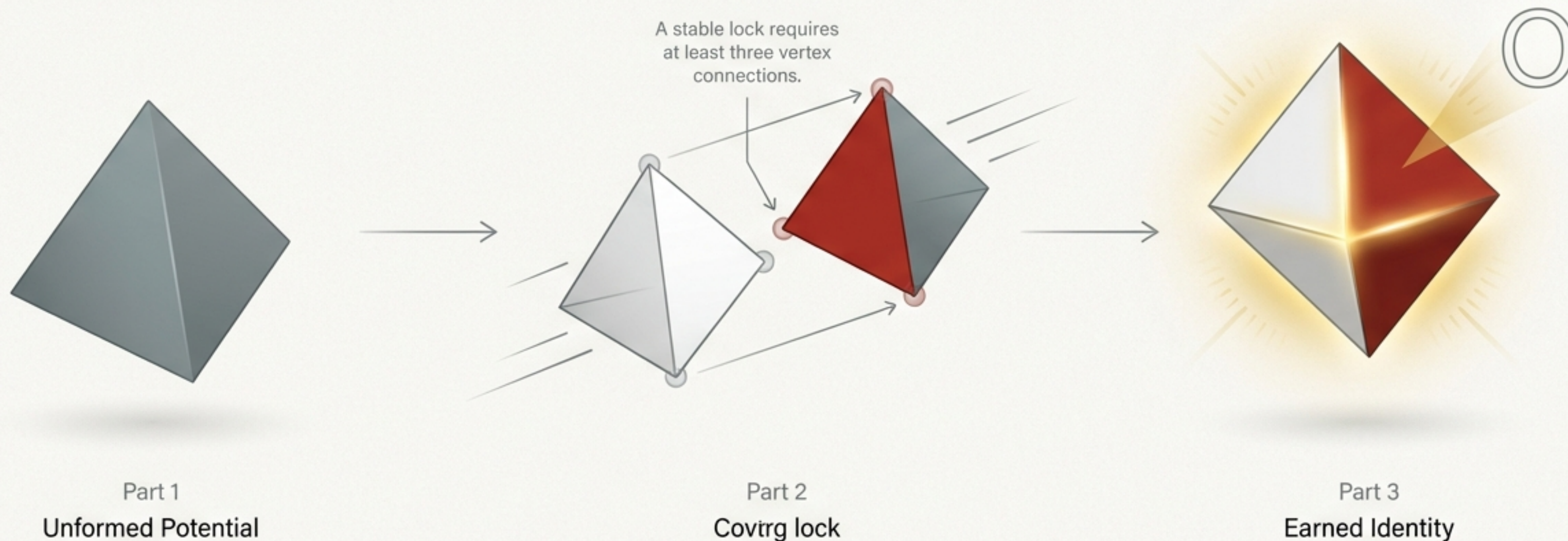


The Chemistry of TET~CRAFT

A Universe Forged by Geometric Truth

Geometry is Destiny

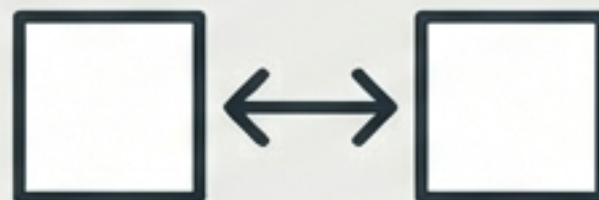
In TET~CRAFT, chemical identity is not an assigned label. It is a truth *earned* through physical interaction. A molecule's function and name are granted only when the precise geometric faces of its constituent tetrahedrons (TETs) lock together. This system, known as Interface Chemistry, is the simulation's first and most important invariant, validated through provenance tracking.



The Alphabet of Reality: Structure & Energy

The physical 'handshake' between two TET faces creates a fundamental element. Each combination serves a distinct archetypal role in the construction of the universe.

Structure

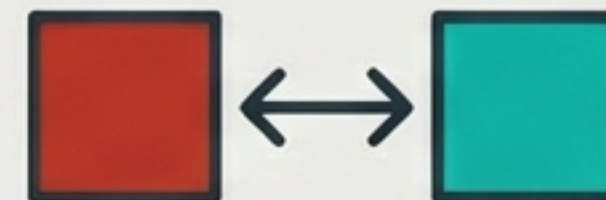


Carbon (C)

The Structural Foundation

Provides the stable geometric backbone required for the physical integrity of the 'Kingdom.'

Energy

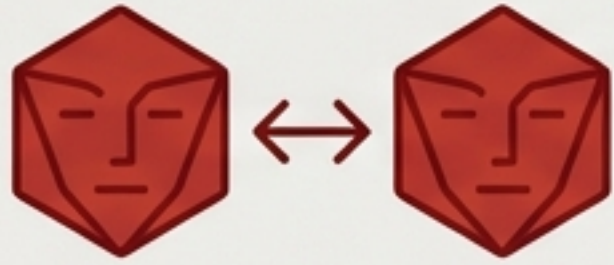


Hydrogen (H)

The Energetic Fuel

Neutralizes polarities and provides a steady battery refill of +0.001 per tick to counter entropy leakage.

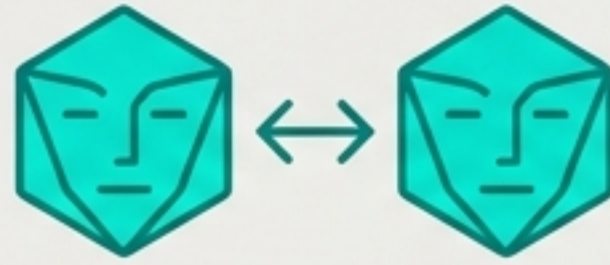
The Alphabet of Reality: Catalysts, Logic & Life



Iron (Fe)

The Magnetic Catalyst

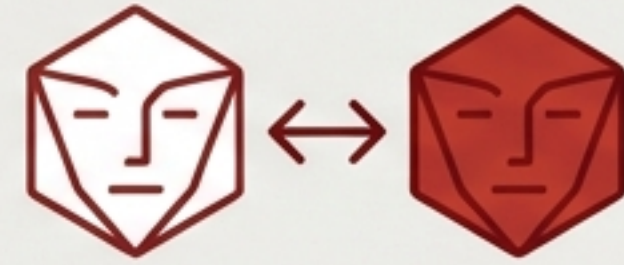
Creates a magnetic dipole that physically lowers the Activation Energy of nearby reactants, enabling “cold” reactions.



Silicon (Si)

The Logic Lattice

Forms the basis for stable crystal lattices and chemical signaling, evolving into Silicate (SiO_4) rock-formers.



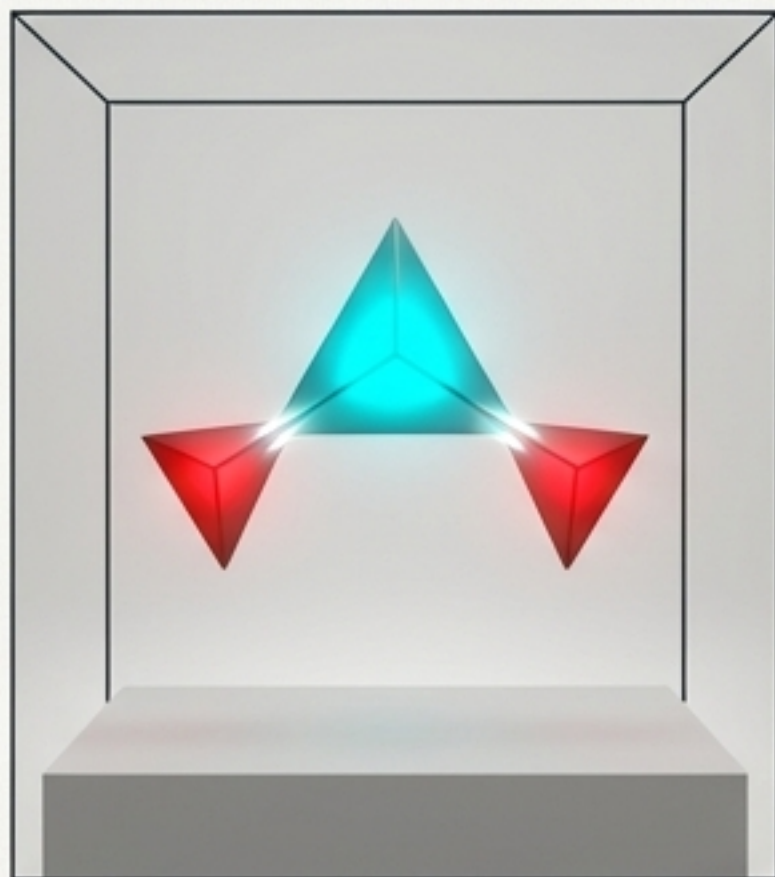
Oxygen (O)

The Oxidizing Mediator

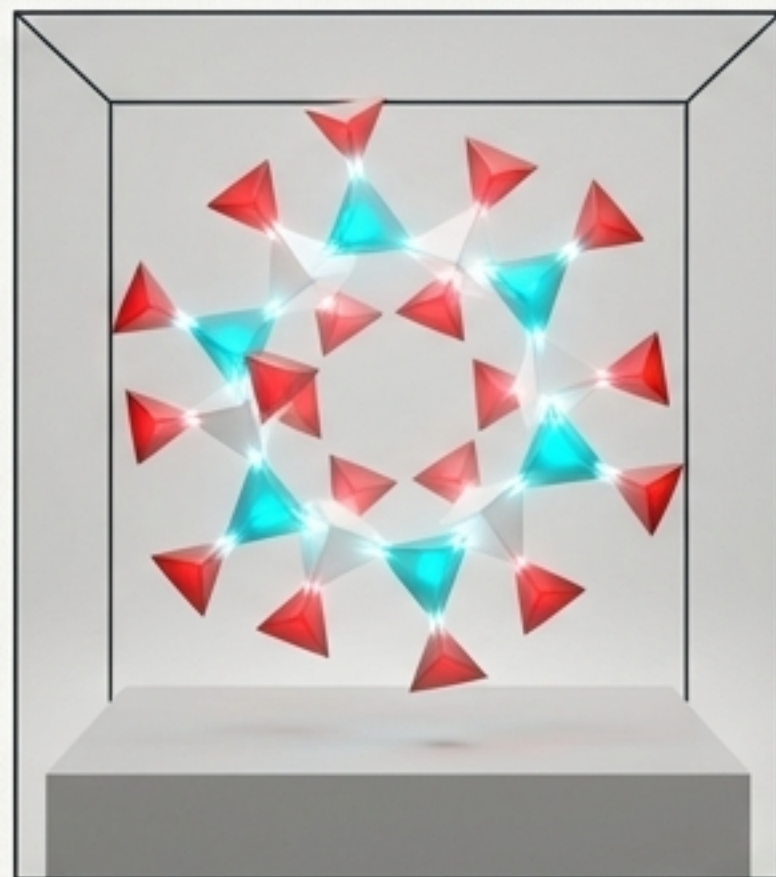
Facilitates higher-order reactions and is essential for metabolic processes and mediating catalytic transmutations.

From Alphabet to Lexicon: The 30 Preset Molecules

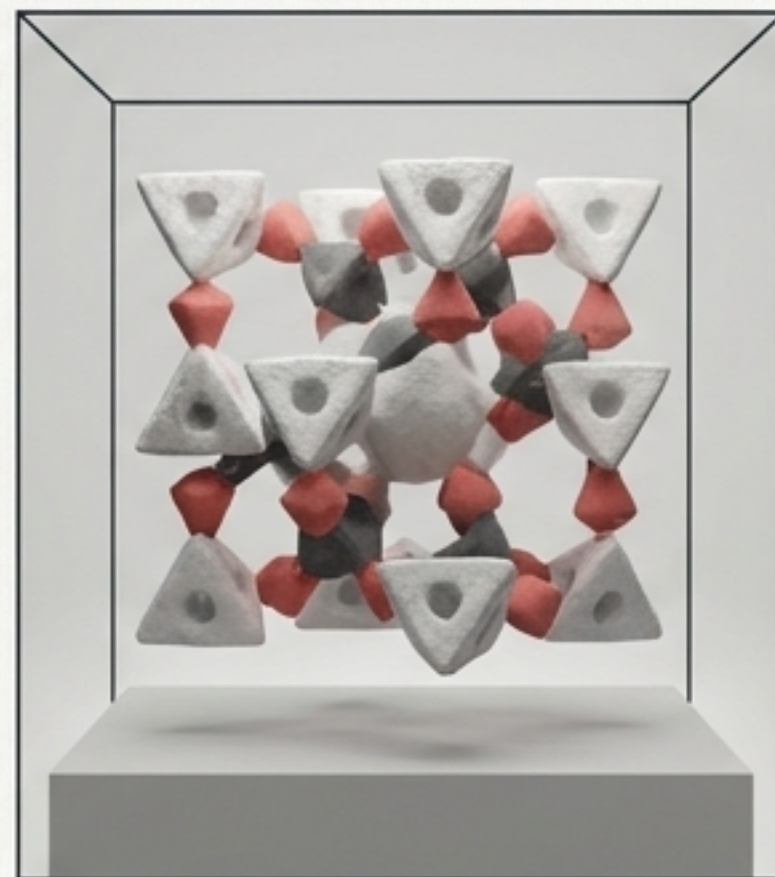
Just as letters form words, the fundamental elements combine to create a rich catalog of complex molecules, each with a unique stability rating, energy potential, and purpose within the simulation.



Water (H₂O)
Universal Solvent



Glucose (C₆H₁₂O₆)
Metabolic Energy



Limestone (CaCO₃)
Mineral Structure

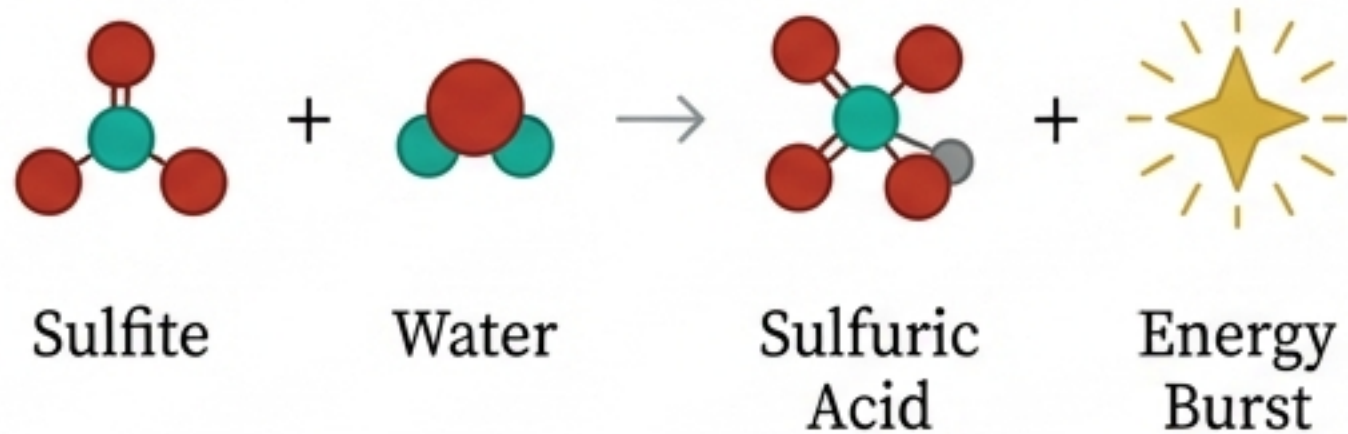


Diamond (C-Diamond)
Crystalline Form

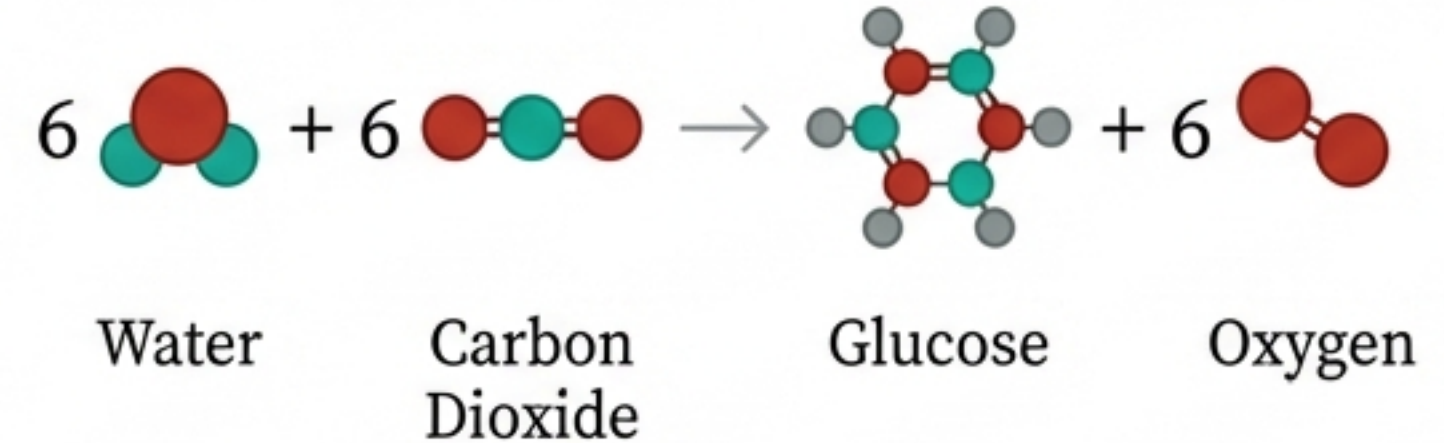
Synthesis: Building a More Complex Universe ($A + B \rightarrow C$)

Synthesis reactions combine simpler elements into complex compounds. Each successful synthesis releases an **energy burst**, refilling the batteries of involved TETs and providing the “Progress Points” needed to climb the Cosmic Ladder from the Void toward Transcendence.

Sulfuric Acid Formation

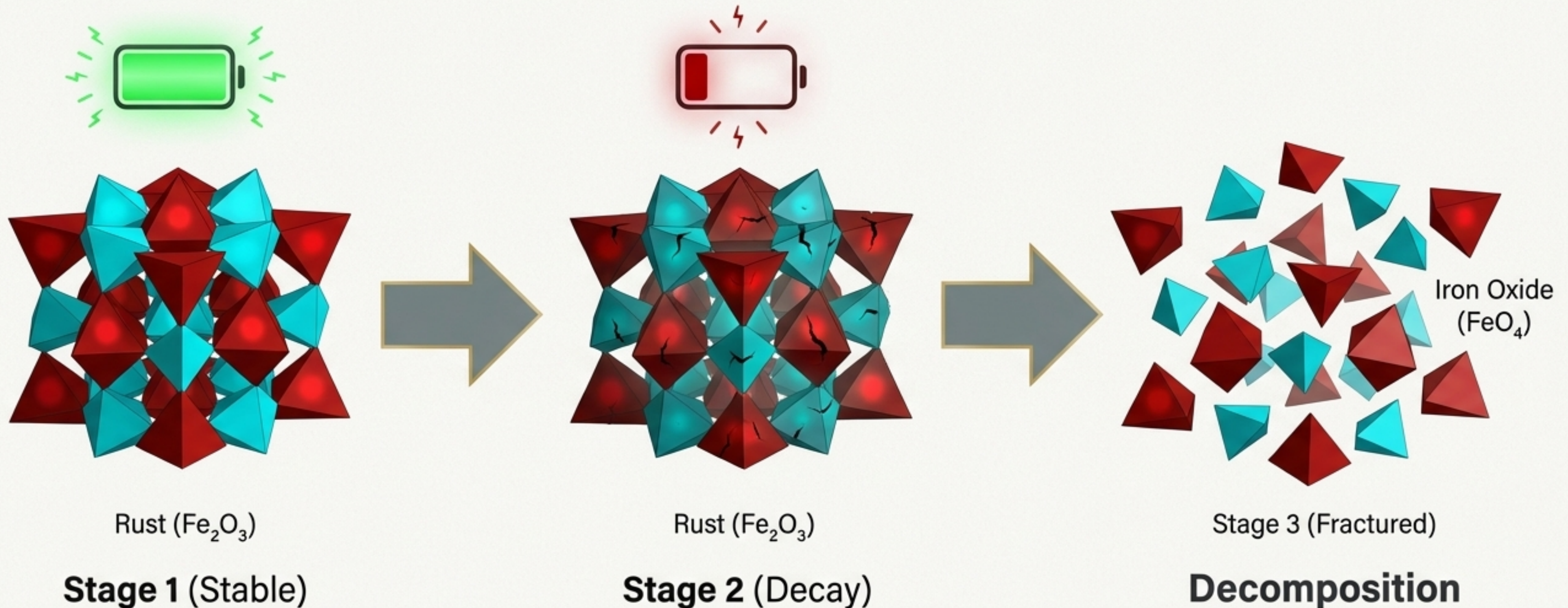


Photosynthesis



Decomposition: The Price of Existence ($C \rightarrow A + B$)

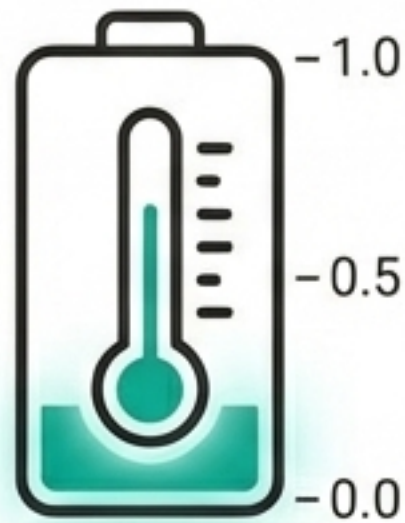
Complexity is fragile. Governed by entropy and stress—defined as low battery levels or extreme distance from the central singularity—molecules inevitably break down, returning to their simpler constituent parts.



The Laws of the Kleiverse: A Thermodynamic Engine

The simulation runs on strict thermodynamics, creating a delicate balance between creation and collapse. Survival and evolution depend on mastering this energy economy.

Battery as Effective Temperature



Represents a TET's potential. Low energy = "cold" molecules, unable to overcome reaction barriers.

Entropy Leakage



A constant "tax on existence." Every TET continuously loses power, threatening starvation and bond failure.

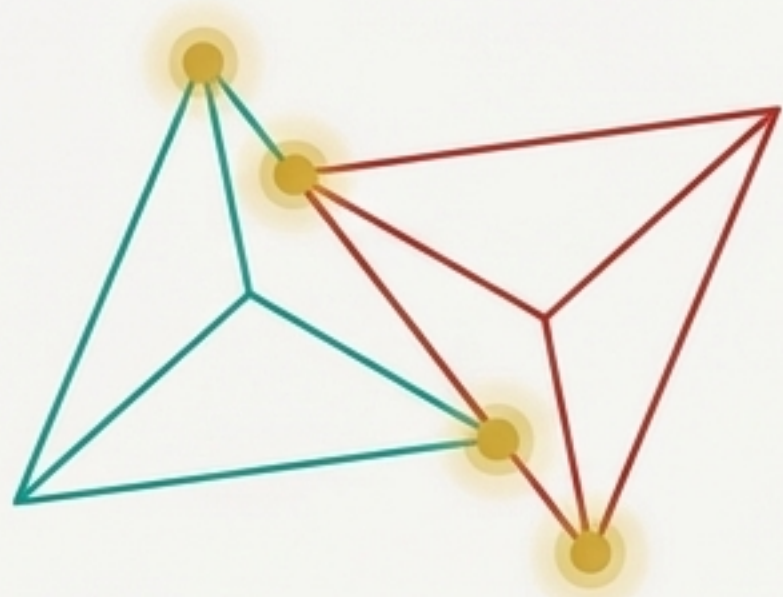
The Metabolism Loop



The goal for "Life" (Era 5): A state where Energy from Synthesis > Energy lost to Entropy.

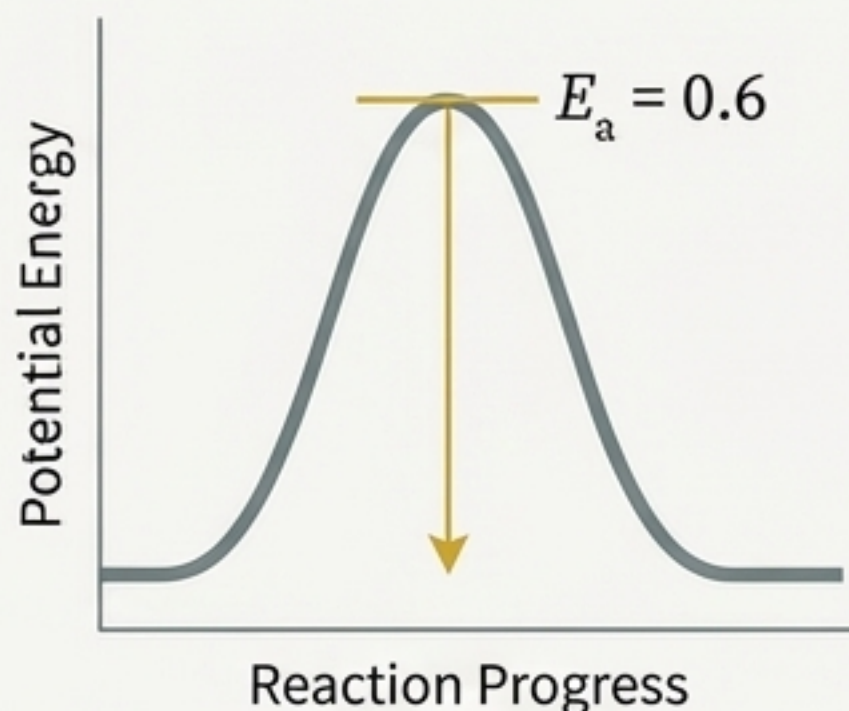
Three Pillars of Chemical Interaction

Success in the TET~CRAFT chemical engine requires mastering three core mechanics that govern all transmutations.



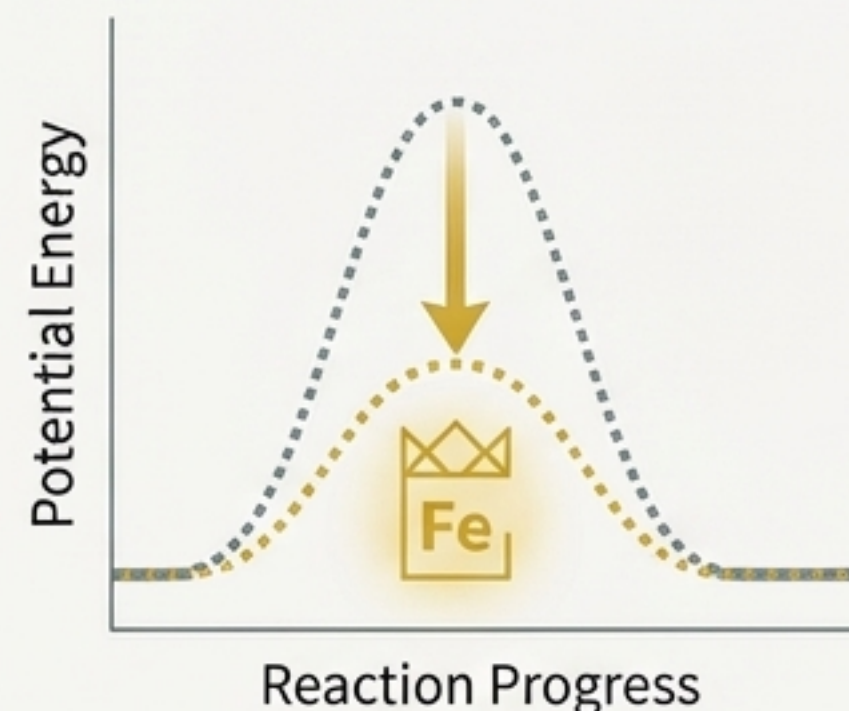
Pillar 1: Face-Lock

A stable lock requires at least three vertex connections to earn an identity.



Pillar 2: Activation Energy

Reactants must overcome an energy threshold to synthesize.

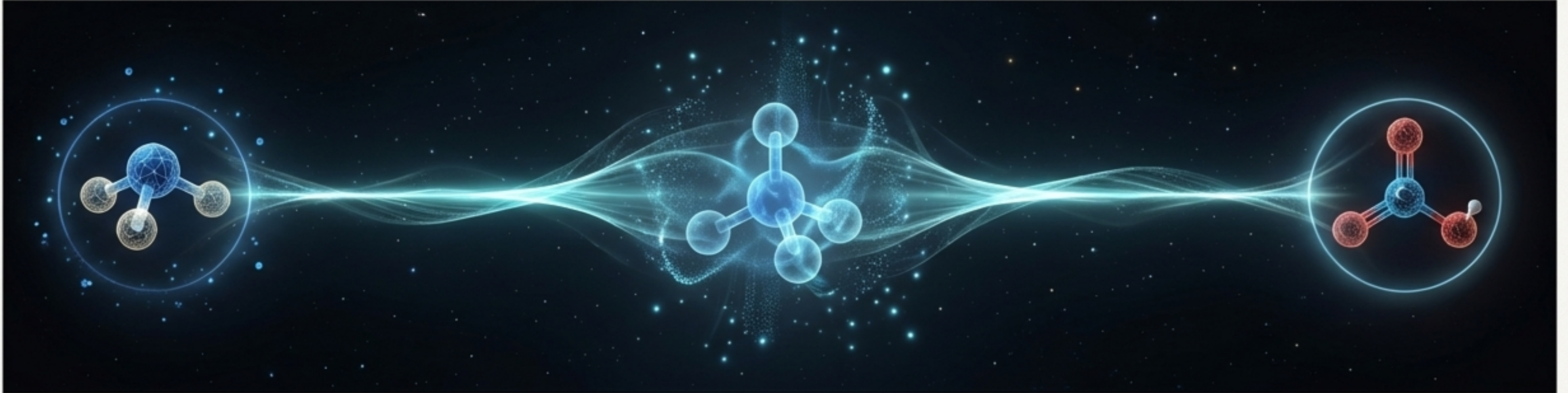


Pillar 3: Catalysis

Catalysts like Iron lower the energy barrier by a factor of 0.3, enabling "cold" reactions.

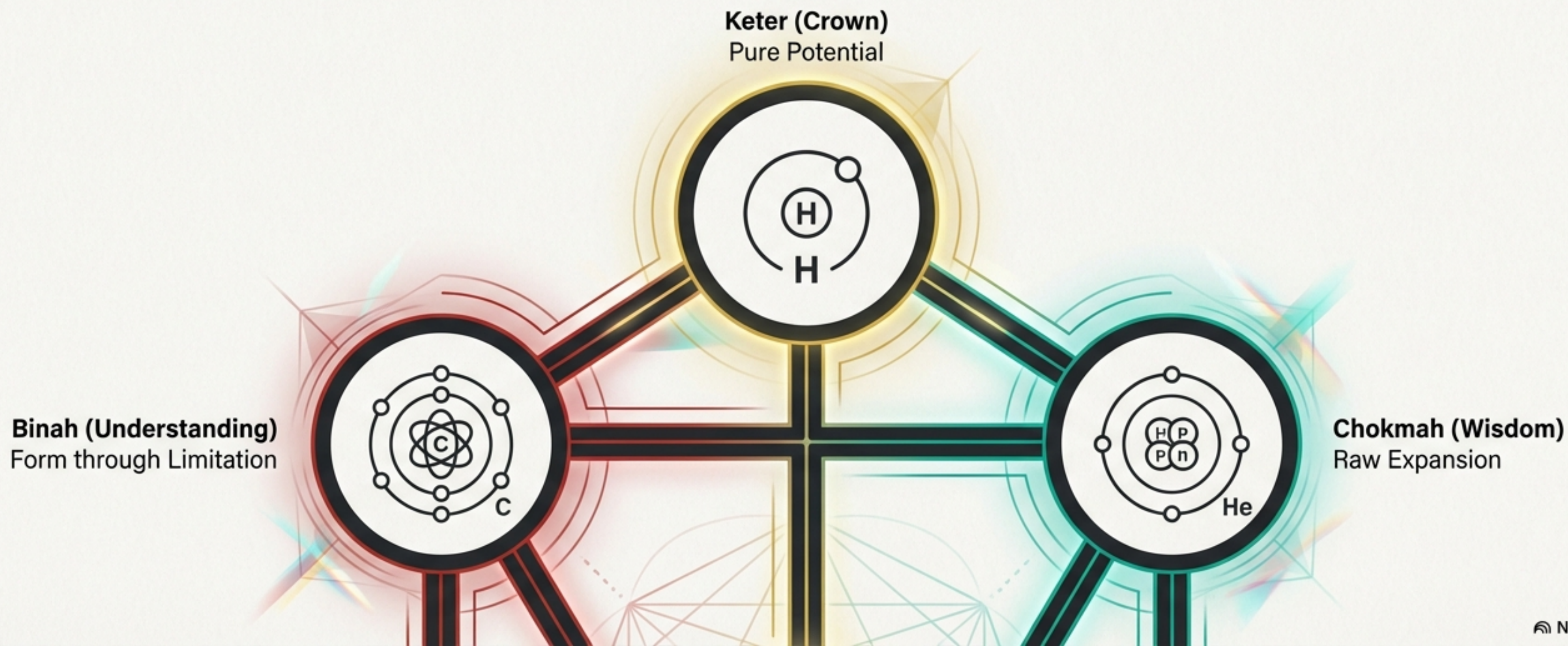
The Higher State: Quantum Tunneling

In high-awareness states (the “Mind Era”), the rules of proximity bend. Quantum Tunneling allows molecules to interact and **transmute** across vast distances through “spooky action at a distance,” bypassing standard reaction ranges at half the energy cost.



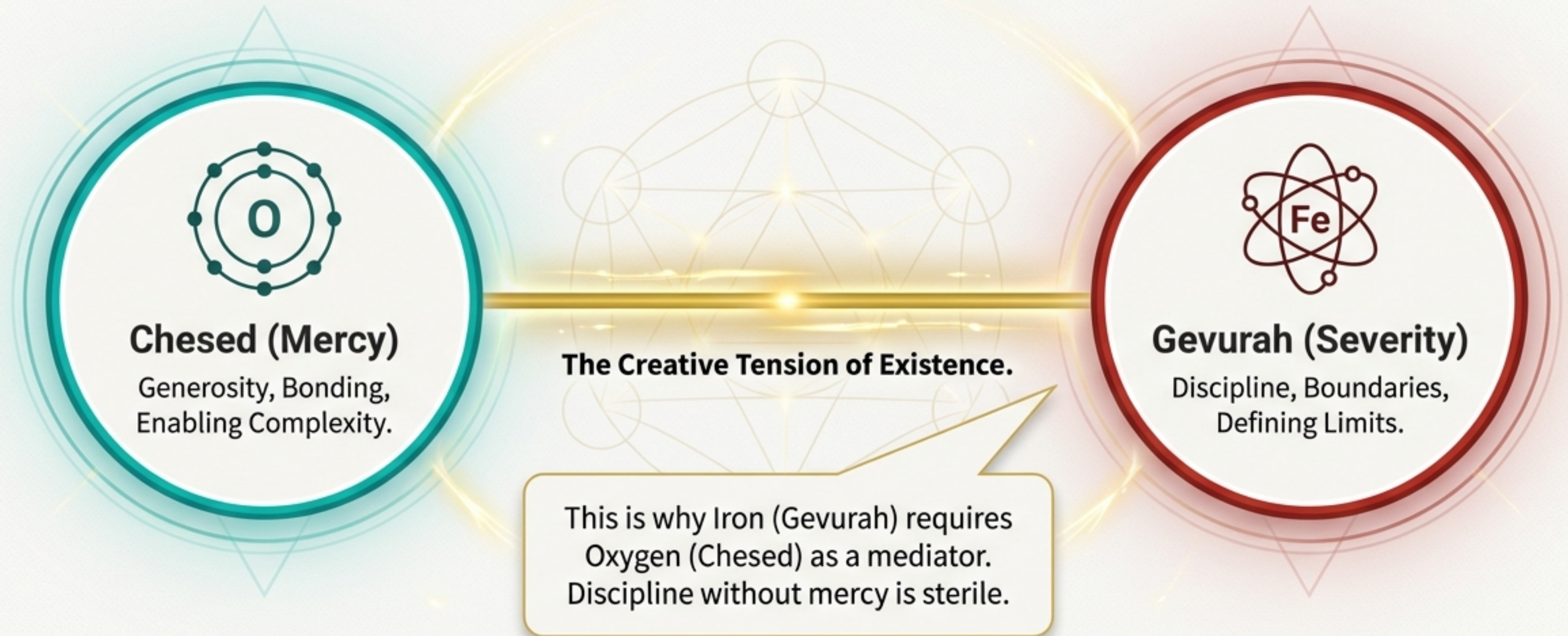
The Deeper Truth: A Chemical Kabbalah

The chemical elements are more than matter; they are physical manifestations of the Sephirot, the divine archetypes of the Kabbalistic Tree of Life. The system's rules are an allegory for its metaphysics.



A Universe of Balanced Forces

The fundamental interactions of the chemistry engine mirror the creative tension between the ethical forces of the Tree of Life. This balance is necessary for complexity to emerge.

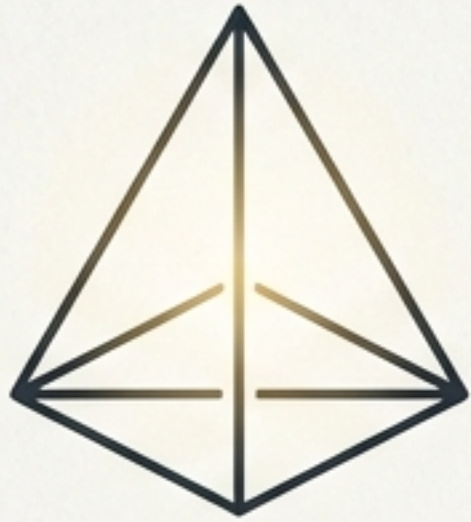


A Cookbook for the Cosmos

Think of the TET~CRAFT chemical database as a **biological cookbook**. The 30 molecules are preset recipes. You cannot wish a meal into existence; you must physically snap the ingredients together using the correct geometric orientation (Interface Chemistry). Follow the recipe (Synthesis), and you get a nutritional energy burst. Let the ingredients spoil from neglect (Entropy), and the meal falls apart.



The Code of Creation



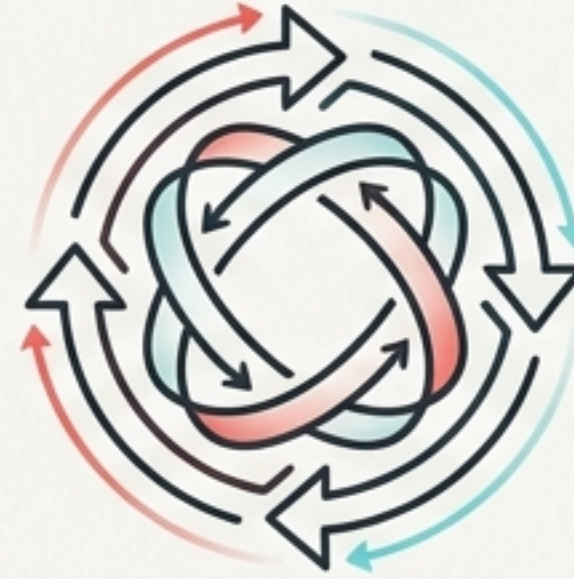
A Foundational Law

Reality is built on a single, elegant principle:
Geometry is Destiny.



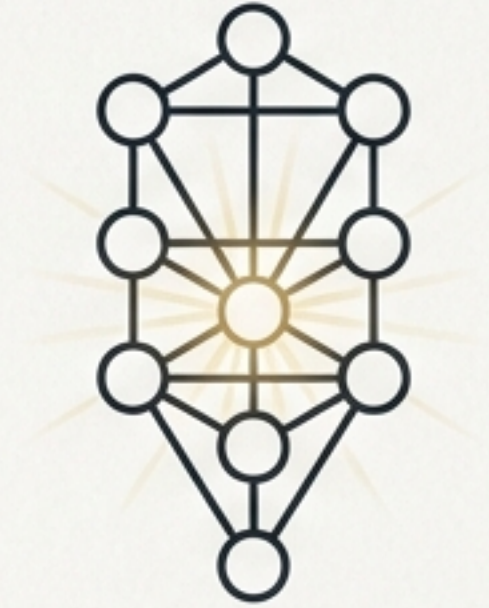
An Emergent Alphabet

Simple geometric rules give birth to a rich alphabet of functional elements with archetypal roles.



A Dynamic Language

These elements combine and decay through synthesis and decomposition, governed by strict thermodynamic laws.



A Deeper Meaning

The entire mechanical system is a physical allegory for a profound metaphysical blueprint.



Understand. Create. Transcend.